

ADD/CHANGE REQUEST FORM

Change Request #CR-2026-001: Title revision, system branding revision, and matching-algorithm enhancement

Project Name: Development of Fira: Web-Based Recruitment Platform for FIL International Recruitment Agency with Applicant Tracking and SBERT-Based Candidate-Job Matching

Requested By: Thesis proponents

Contact:
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Date of Request: September 18, 2026

Description of Add/Change:

- Title revised from “Development of Fira: Web-Based Recruitment Platform for FIL International Recruitment Agency with Applicant Tracking and SBERT-Based Candidate-Job Matching” to “Development of MatchWise: Recruitment and Applicant Management System Utilizing SBERT, Cosine Similarity, and LightGBM for Applicant-Job Matching at FIL International Recruitment Agency”
- “Replace the system codename “FIRA” with “MatchWise” throughout the project
- Revise the matching algorithm from the current “SBERT embeddings + cosine similarity” approach to an extended “SBERT embeddings + cosine similarity + LightGBM-based ranking” approach.
- Clarify the AI matching terminology by identifying:
 - SBERT as the embedding model;
 - Cosine Similarity as the similarity measure used to compare embeddings; and
 - LightGBM as the learning-to-rank model that combines similarity and structured recruitment features.

Reason for Add/Change:

- To resolve a naming conflict between the working system name "FIRA" and the client's own acronym (FIL International Recruitment Agency), and to strengthen the objectivity of candidate-job matching by supplementing SBERT embedding similarity with a LightGBM learning-to-rank model that also considers structured recruitment features (e.g., required skills, experience level, location), as recommended during panel/adviser review of the proposal.

Change Impact

Tasks/Scope Affected:

- Revise thesis manuscript (title page, Chapters 1, 2, 3) to reflect the new title and system name; update UI labels, source code identifiers, database references, and system diagrams from "FIRA" to "MatchWise"; extend the AI matching microservice to add a LightGBM ranking component on top of the existing SBERT + Cosine Similarity pipeline; and update the benchmarking and methodology sections accordingly.

Risk Evaluation:

- Low to moderate risk. The additional LightGBM ranking component increases implementation and testing scope, but the existing SBERT + Cosine Similarity approach remains functional as a baseline; thus, the system is not left without a working matching mechanism at any point during the transition.

Quality Evaluation:

- Expected to improve overall system quality: matching results should be more accurate since ranking will account for structured recruitment features in addition to semantic similarity, and the clarified title/terminology reduces ambiguity between the client's name and the system's name in the documentation.

Additional Resources:

- Historical or sample applicant-job data with outcome labels (e.g., shortlisted/hired) for training and validating the LightGBM model; Python libraries LightGBM and scikit-learn added to the AI microservice environment; and additional development/testing time from the team member(s) handling the matching module.

Duration:

- Approximately 3–4 weeks: LightGBM model development, training, and evaluation, together with rebranding and manuscript revision tasks carried out in parallel.

Impact Assessment

Timeline Impact

- Adds an estimated 3–4 weeks to the development schedule for training and validating the LightGBM ranking model; rebranding and documentation updates can proceed in parallel and are not expected to delay the overall thesis timeline.

Budget Impact

- None. LightGBM and its supporting libraries are open-source and compatible with the existing Python/FastAPI-based AI microservice, so no additional cost is incurred.

Quality Impact

- Positive. Combining SBERT/Cosine Similarity with LightGBM-based ranking is expected to improve match relevance, and clarifying the SBERT/Cosine Similarity/LightGBM roles improves the technical clarity of the documentation.

Resource Impact

- Additional effort is required from the member(s) responsible for the AI matching microservice to implement and evaluate the LightGBM model, and from the rest of the group to carry out the "FIRA"-to-"MatchWise" rebranding across code, database, diagrams, and manuscript.

Risk Impact

- Low. Because the SBERT + Cosine Similarity approach is retained as the underlying similarity measure and LightGBM is added as a ranking layer rather than a replacement, the risk of losing existing, already-working functionality is minimal.

Comments:

This change does not alter the core system architecture or user roles already approved in Chapter 3; it refines the title/terminology and extends the matching pipeline previously presented to the panel.

Sign Offs

[Circle One]: 1. Accepted 2. Deferred 3. Rejected 4. More Info Requested

Comments:

Course Adviser's Signature:

Date:

Client's Signature:

Date: